# HAN ZHANG

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## WORKING EXPERIENCE

**Brown University, Rhode Island** — *August 2026 - Ongoing*
Division of Applied Mathematics — Postdoctoral Research Associate
Work with George Karniadakis,

## EDUCATION

**City University of Hong Kong, Hong Kong** — *July 2026*
Department of Mathematics — Ph.D. in Mathematics
Supervised by Raymond H. CHAN,
Co-supervised by Xue-Cheng TAI and Jean-Michel MOREL

**Chinese University of Hong Kong, Hong Kong** — *July 2020*
Department of Mathematics — M.Phil. in Mathematics
Supervised by Lok Ming LUI

**Sun Yat-Sen University, Guangdong** — *June 2018*
School of Mathematics — B.Sc. in Computational Science

## RESEARCH INTEREST

**Scientific Computing**: fluid-structure interaction, inverse problems
**Scientific Machine Learning**: physics-informed learning, neural operators
**Computational Geometry**: quasi-conformal geometry, deformable domains
**Applications**: cardiovascular modeling, blood flow simulation, and imaging science

## VISITING SCHOLAR

**Emory University:** Hosted by Yuanzhe Xi and Alessandro Veneziani. *Oct. 2025*
**Princeton University:** Hosted by Guillermo Sapiro. *May 2025 — Oct. 2025*
**Shenzhen University:** Hosted by Hui Huang. *Aug 2020 — July. 2021*

## JOURNAL PUBLICATIONS

$^{\mathsf{T}}$denotes the corresponding author.
$^{*}$denotes the equal contribution.

1. Reconstructing Interacting Blood Flow and Elastic Vessel-Wall Deformation from Noisy Flow Data Using PirateNets.
   **Han Zhang**$^{\mathsf{T}}$, Xue-Cheng Tai, Alessandro Veneziani, Raymond H. Chan
   Submitted to *SIAM Journal of Scientific Computing (SISC).*

2. Learning Task-Aware Sampling with Shared Saliency through Density-Equalizing Mappings.
   Tsz Lok Ip, **Han Zhang**, Lok Ming Lui$^{\mathsf{T}}$
   Submitted to *Journal of Mathematical Imaging and Vision (JMIV).*

3. A Bifurcation-Based Domain Decomposition Method with Neural Operators for Blood Flow Simulation .
   Yuzhou Zhao, **Han Zhang**$^{\mathsf{T}}$, J. Matias Di Martino, Jean-Michel Morel, Guillermo Sapiro
   *SIAM Multiscale Modeling and Simulation (SIMMS).*

4. Coupled Reconstruction of 2D Blood Flow and Vessel Geometry from Noisy Images via Physics-Informed Neural Networks and Quasi-Conformal Mapping.
   **Han Zhang**$^{\mathsf{T}}$, Xue-Cheng Tai, Jean-Michel Morel, Raymond H. Chan
   *SIAM Journal of Imaging Science (SIIS), 2026.*

5. Circular quasiconformal deturbulence: Geometry-based restoration from multiple turbulent frames.
   Chu Chen, **Han Zhang**, Lok Ming Lui$^{\mathsf{T}}$
   *Neurocomputing (NEUCOMP)*

6. Quasi-Conformal Convolution : A Learnable Convolution for Deep Learning on Simply Connected Open Surfaces.
   **Han Zhang**, Tsz Lok Ip, Lok Ming Lui$^{\mathsf{T}}$
   *SIAM Journal of Imaging Science (SIIS), 2026.*

7. Parametrized Sampling for 3D Blood Simulation in Deformable Vessels Using Physics-Informed Neural Networks.
   **Han Zhang**, Lingfeng Li, Xue-Cheng Tai$^{\mathsf{T}}$, Raymond H. Chan
   *Journal of Computational and Applied Mathematics (JCAM), 2025.*

8. Deformation-Invariant Neural Network and Its Applications on Image Classification and Restoration.
   **Han Zhang**, Qiguang Chen, Lok Ming Lui$^{\mathsf{T}}$
   *Neural Network (NEU NET), 2025.*

9. QIS : Interactive Segmentation via Quasi-Conformal Mappings.
   **Han Zhang**, Daoping Zhang, Lok Ming Lui$^{\mathsf{T}}$
   *SIAM Journal of Imaging Science (SIIS), 2025.*

10. Full 3D Blood Flow Simulation in Curved Deformable Vessels Using Conditional Physics-Informed Neural Networks.
    **Han Zhang**, Xue-Cheng Tai$^{\mathsf{T}}$
    *Acta Mathematica Universitatis Comenianae (AMUC), 2024.*

11. A Meshless Solver for Blood Flow Simulations in Elastic Vessels Using Physics-Informed Neural Network.
    **Han Zhang**, Raymond H. Chan, Xue-Cheng Tai$^{\mathsf{T}}$
    *SIAM Journal of Scientific Computing (SISC), 2024.*

12. A Learning-based Framework for Topology-Preserving Segmentation using Quasiconformal Mappings.
    **Han Zhang**, Lok Ming Lui$^{\mathsf{T}}$
    *Neurocomputing (NEUCOMP), 2024.*

13. A New Framework for Deformation Analysis with Uncertainties using Beltrami Fourier Descriptor.
    **Han Zhang**, Lok Ming LUI$^{\mathsf{T}}$.

*Mathematics, Computation and Geometry of Data (MCGD), 2022.*

14. Quasi-Conformal Neural Network (QC-net) with Applications to Shape Matching.
**Han Zhang**, Lok Ming LUI$^{\mathsf{T}}$.
*Mathematics, Computation and Geometry of Data (MCGD), 2021.*

15. Continuous Aerial Path Planning for 3D Urban Scene Reconstruction.
**Han Zhang**, Yucong Yao, Ke Xie, Chi-Wing Fu, Hao Zhang, Hui Huang$^{\mathsf{T}}$.
*ACM Transaction on Computer Graphics (ACM TOG, SIGGRAPH ASIA), 2021.*

## PROCEEDING PUBLICATIONS

1. Fast Physics-Informed Learning via Diffusion Hypernetworks.
Yuzhou Zhao, **Han Zhang**$^{\mathsf{T}}$, J. Matias Di Martino, Jean-Michel Morel, Guillermo Sapiro
Submitted

2. Nondeterministic Deformation analysis using Quasiconformal Geometry.
**Han Zhang**, Lok Ming Lui$^{\mathsf{T}}$
*IEEE International Conference on Image Processing (ICIP), 2022.*

## ACADEMIC ACHIEVEMENTS

**Outstanding Academic Performance Award, 2024**
**Excellent Student Scholarship of Sun Yat-Sen University, 2017** *First Class*
**Excellent Thesis of Sun Yat-Sen University, 2018** *Outstanding*
**China Undergraduate Mathematical Contest in Modeling, 2016** *Second Prize*
**National High School Mathematics League, 2012** *Second Prize*

## INVITED TALKS

1. [**1-hour Invited Talk**] Physics-Informed Neural Network for Blood Flow Simulation : Forward and Inverse Problem
Tsinghua University - MATH+ML+X Seminar, Online *May 2026*
2. [**1-hour Invited Talk**] Physics-Informed Neural Network for Blood Flow Simulation : Forward and Inverse Problem
Brown University - CRUNCH Seminar, Online *Mar. 2026*
3. [**1-hour Invited Talk**] Physics-Informed Neural Network for Blood Flow Simulation : Forward and Inverse Problem
Emory University - CODES Seminar, Atlanta, USA *Oct. 2025*

## REVIEW

1. Computer Graphics Forum
2. Journal of Mathematical Imaging and Vision
3. Journal of Mathematical Learning and Computation
4. Medical Image Analysis
5. Neural Networks
6. Neurocomputing
7. Pattern Recognition
8. Physics of Fluids

## TEACHING EXPERIENCE

* denotes courses with duty of a weekly one-hour instructional session.

**City University of Hong Kong**

1. MA1200 Calculus & Basic Linear Algebra I* *Spring, 2025*
2. MA1201 Calculus & Basic Linear Algebra II* *Spring, 2025*
3. MA2177 Engineering Mathematics and Statistics* *Fall, 2024*
4. MA1200 Calculus & Basic Linear Algebra I* *Spring, 2024*
5. MA1200 Calculus & Basic Linear Algebra I* *Fall, 2023*
6. GE1359/MA1502 Algebra *Spring, 2023*

**Chinese University of Hong Kong**

1. MATH1010 University Mathematics *Spring, 2020*
2. MATH1510B Calculus for Engineers *Fall, 2019*
3. MATH4250 Game Theory *Spring, 2019*
4. MATH1510B Calculus for Engineers *Fall, 2018*